PENETRATION TESTING REPORT

FOOTPRINTING, RECONNAISSANCE & NETWORK SCANNING

WEEK 2 | CYBERSECURITY INTERNSHIP | NETWORKWALKS

Report Detail	Information
Trainee Name	Hadiya Khan
Program / Batch	Networkwalks – B084
Report Date	11 November 2026
Modules Completed	Module 1 – Multiple Kali Tools; Module 4 – theHarvester; Module 5 – Zenmap
Footprinting Target	github.com
Local Network Scope	Own Wi-Fi LAN – 192.168.0.0/24
Platforms	Kali Linux and Windows 11

Prepared as part of the Week 2 practical activities

1. Liability Disclaimer

The activities documented in this report were performed for educational purposes as part of the Networkwalks cybersecurity internship. Domain reconnaissance was limited to publicly accessible information about github.com. Network discovery was performed on the trainee's own local Wi-Fi network, 192.168.0.0/24.

These exercises did not include exploitation, service intrusion, credential attacks, or modification of target systems. Findings are observations at the time of testing and are not proof of vulnerabilities. Any future security testing must be conducted only with appropriate authorization and within an explicitly approved scope.

2. Introduction

This report documents three Week 2 practical modules: Module 1 (Footprinting and Reconnaissance with Multiple Kali Tools), Module 4 (Footprinting with theHarvester), and Module 5 (Network Scanning with Zenmap). The theHarvester practical is labelled Module 4 in the supplied Networkwalks assignment material.

For Modules 1 and 4, github.com was used as the reconnaissance domain. Public domain, DNS, HTTP, web-technology and host-enumeration information was collected. For Module 5, Zenmap was used from a Windows PC to discover active hosts on the trainee's own Wi-Fi subnet. The report records commands, observed outputs, interpretation, limitations and supporting screenshots.

3. Tools Used

Tool / Platform	Purpose
Kali Linux / Windows 11	Operating systems used for reconnaissance and network discovery.
WHOIS	Retrieve public domain registration and registrar information.
WhatWeb	Identify visible web technologies and HTTP characteristics.
nslookup	Resolve a domain name through DNS.
curl -I	Retrieve HTTP response headers.
WAFW00F	Check whether a web application firewall is detected.
DNSRecon	Enumerate DNS records such as NS, SOA, MX, A and TXT.
theHarvester 4.10.1	Collect public host, IP, email, ASN and URL information from configured sources.
Zenmap / Nmap 7.993	Discover live hosts on the local subnet.
Windows CMD	Identify local IP configuration and Wi-Fi adapter MAC address.

4. Activities Performed

4.1 Module 1 – Footprinting & Reconnaissance with Multiple Kali Tools

Target domain: github.com. Six Kali Linux tools were used to collect different categories of publicly available information.

4.1.1 WHOIS – Domain Registration Details

Command: `whois github.com`

The WHOIS output identified MarkMonitor Inc. as registrar. The domain creation date shown was 9 October 2007 and the registry expiry date was 9 October 2028. Eight name servers were listed across NSone and Amazon Route 53 naming domains. Registrant details were privacy-redacted. The record showed clientUpdateProhibited, clientTransferProhibited and clientDeleteProhibited statuses.



Figure 1. Module 1 – WHOIS domain registration output.

4.1.2 WhatWeb – Web Technology Fingerprinting

Command: `whatweb github.com`

WhatWeb reported an HTTP 301 redirect from HTTP to HTTPS, followed by 200 OK. The output identified IP address 20.207.73.82, HTML5, server label github.com, page-title information, cookies and response headers including Strict-Transport-Security, X-Frame-Options and X-XSS-Protection. This is visible fingerprint information, not proof of a weakness.

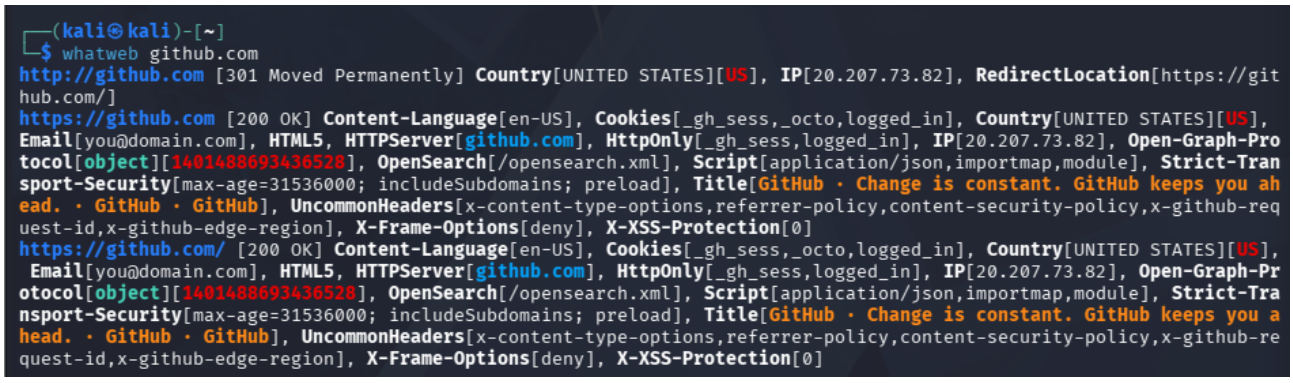


Figure 2. Module 1 – WhatWeb fingerprinting output.

4.1.3 nslookup – DNS Resolution

Command: `nslookup github.com`

The query used DNS server 8.8.8.8 and returned a non-authoritative answer resolving github.com to 20.207.73.82.

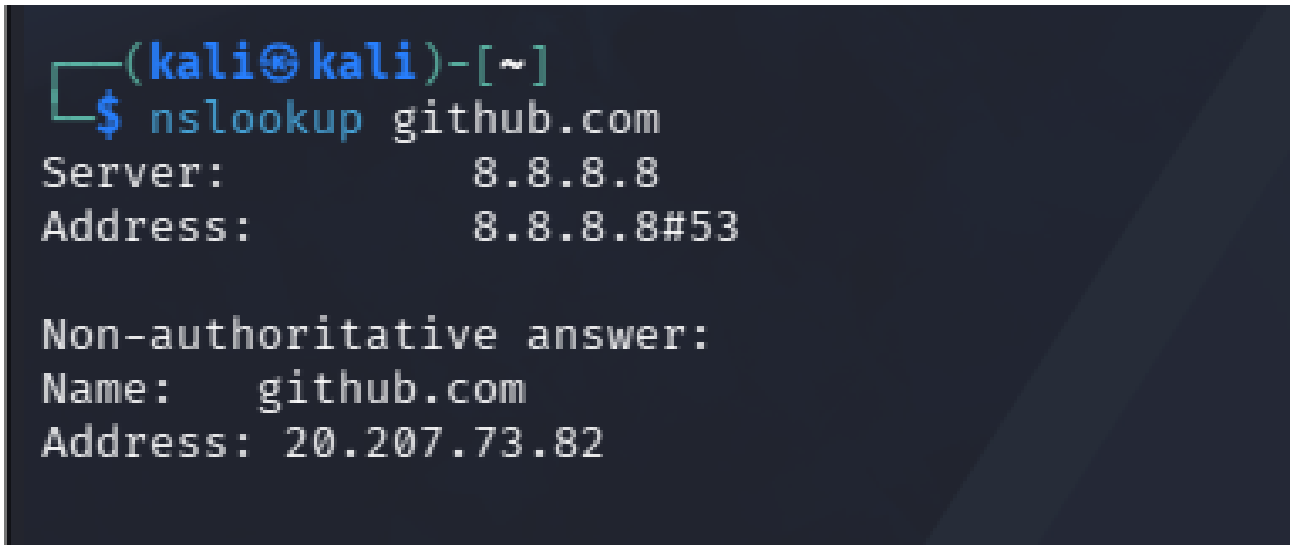


Figure 3. Module 1 – nslookup DNS resolution output.

4.1.4 curl – HTTP Response Headers

Command: `curl -I https://github.com`

The server returned HTTP/2 200. Headers included content-type text/html; charset=utf-8, content-language en-US, cache-control directives, Strict-Transport-Security, X-Frame-Options: deny, X-Content-Type-Options: nosniff, Referrer-Policy and Content-Security-Policy. These headers describe the server response and browser policies.

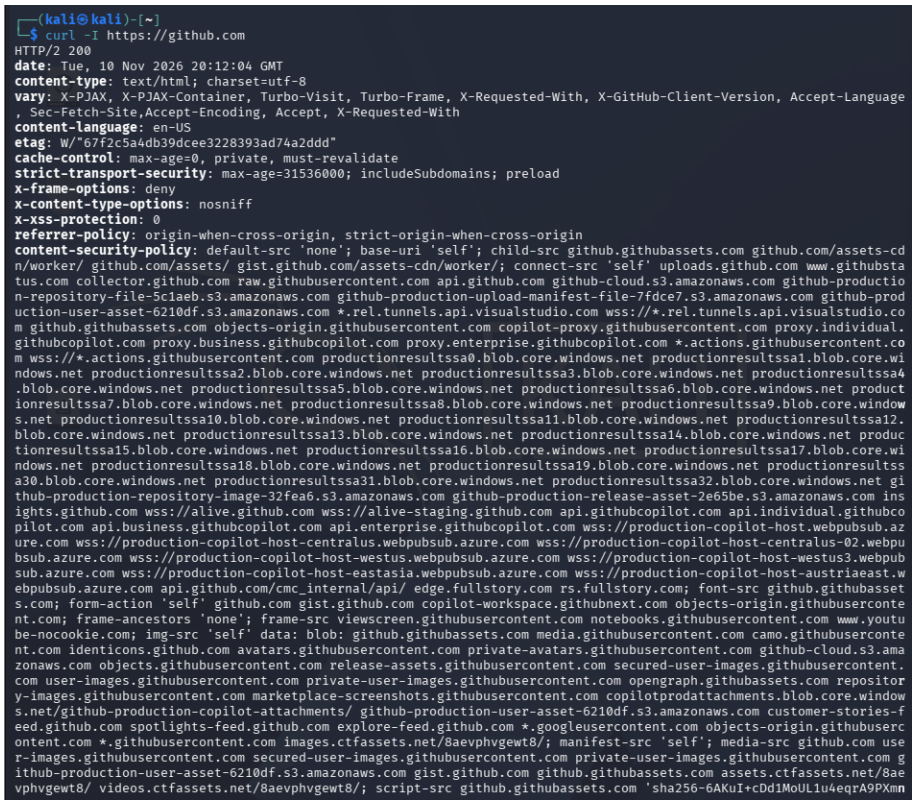


Figure 4. Module 1 – curl HTTP response headers.

4.1.5 WAFW00F – Web Application Firewall Detection

Command: `wafw00f github.com`

WAFW00F v2.4.2 reported: “No WAF detected by the generic detection.” The output recorded seven requests. This means the generic detection did not identify a WAF; it does not prove that no firewall or other protective control exists.



Figure 5. Module 1 – WAFW00F detection output.

4.1.6 DNSRecon – DNS Enumeration

Command: `dnsrecon -d github.com`

DNSRecon reported SOA, NS, MX, A and TXT information. The A record shown for github.com was 20.207.73.82. The output listed multiple name servers and the mail exchanger github-com.mail.protection.outlook.com. Several TXT records included SPF and domain/service verification data. The tool also logged no answer for a DNSSEC query; this is recorded as an observation, not a confirmed DNSSEC vulnerability.

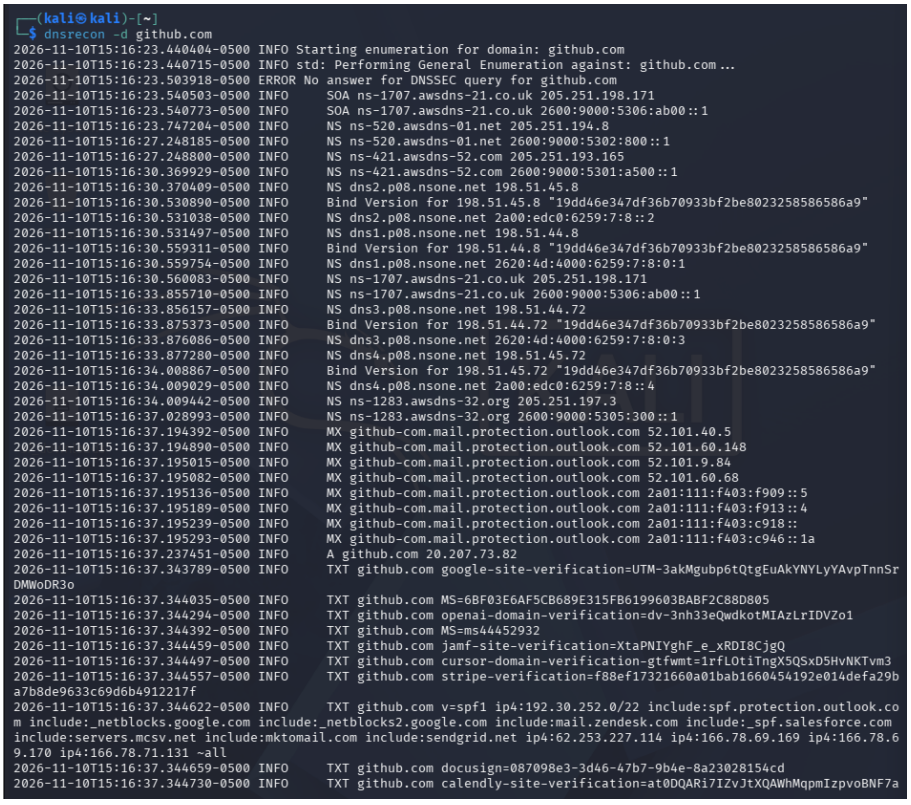


Figure 6. Module 1 – DNSRecon enumeration output.

4.2 Module 4 – Footprinting with theHarvester

Target domain: github.com. Tool version shown in the terminal output: theHarvester 4.10.1.

4.2.1 Task 1 – Hackertarget Source

Command: `theHarvester -d github.com -l 1000 -b hackertarget`

The run reported no IP addresses, no email addresses and no people. It identified 295 hosts. The output included hostnames and associated IP addresses for some entries.



Figure 7. Module 4 – theHarvester results using the Hackertarget source.

4.2.2 Task 2 – All Sources

Command: `theHarvester -d github.com -l 50 -b all`

The all-sources run reported 64 IP addresses, no email addresses and no people, 4,680 hosts, 11 autonomous system numbers (ASNs), and 23 interesting URLs.

Limitations: Several providers returned missing API-key, invalid-credential, authentication or unavailable-service messages. Examples included missing keys for SecurityScorecard, BuiltWith, Shodan, Tomba, VirusTotal, WhoisXML and ZoomEye; invalid Fofa credentials; an incorrect LeakIX API key; and no response from ThreatCrowd. Windvane reported limited unauthenticated access and used a DNS fallback.

Interpretation: theHarvester aggregates information from configured public sources, but completeness depends on provider availability, credentials and service limits. These totals represent the results of this run, not a complete inventory of all GitHub-related assets.



Figure 8. Module 4 – theHarvester all-sources execution and provider responses.

4.3 Module 5 – Network Scanning with Zenmap

Scan scope: the trainee's own Wi-Fi LAN, 192.168.0.0/24. Zenmap was installed on the Windows PC and the Ping Scan profile was selected.

4.3.1 Local IP Address and Subnet

Command: `ipconfig`

The active Wi-Fi adapter showed IPv4 address 192.168.0.102, subnet mask 255.255.255.0 and default gateway 192.168.0.1. The subnet was therefore 192.168.0.0/24. Windows also displayed an Ethernet 3 adapter with 192.168.56.1 and no default gateway; this adapter was not used as the Wi-Fi scan scope.

```
C:\Users\khhad>ipconfig

Windows IP Configuration


Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::471e:d8c5:d2cc:80f6%9
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::dbdf:9c87:5ed8:5532%17
   IPv4 Address. . . . . . . . . . . : 192.168.0.102
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1
```

Figure 9. Module 5 – Windows ipconfig output showing the Wi-Fi network configuration.

4.3.2 Ping Scan and Live Host Discovery

Command: `nmap -sn 192.168.0.0/24`

Zenmap scanned 256 IP addresses and reported six hosts up: 192.168.0.1, 192.168.0.100, 192.168.0.101, 192.168.0.102, 192.168.0.104 (host.docker.internal), and 192.168.0.107. The scan completed in approximately 7.59 seconds. Because -sn performs host discovery without a port scan, these results do not establish which TCP or UDP services are open.

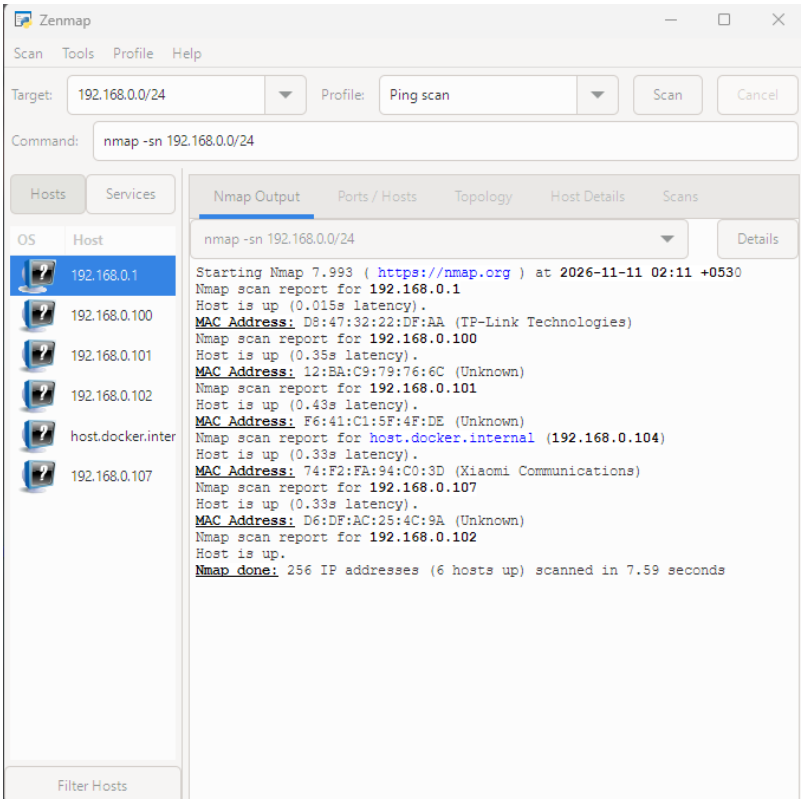


Figure 10. Module 5 – Zenmap output showing six live hosts on 192.168.0.0/24.

4.3.3 MAC Address Identification

Zenmap displayed MAC addresses for five discovered hosts. The MAC address for the Windows Wi-Fi adapter (192.168.0.102) was obtained separately using ipconfig /all.

IP Address	MAC Address	Identification shown
192.168.0.1	D8:47:32:2D:FA:AA	TP-Link Technologies
192.168.0.100	12:BA:C9:79:76:6C	Unknown
192.168.0.101	F6:41:C1:5F:4F:DE	Unknown
192.168.0.104	74:F2:FA:94:C0:3D	Xiaomi Communications; host.docker.internal
192.168.0.107	D6:DF:AC:25:4C:9A	Unknown
192.168.0.102	F8:54:F6:3B:6E:73	Windows Wi-Fi adapter; locally verified

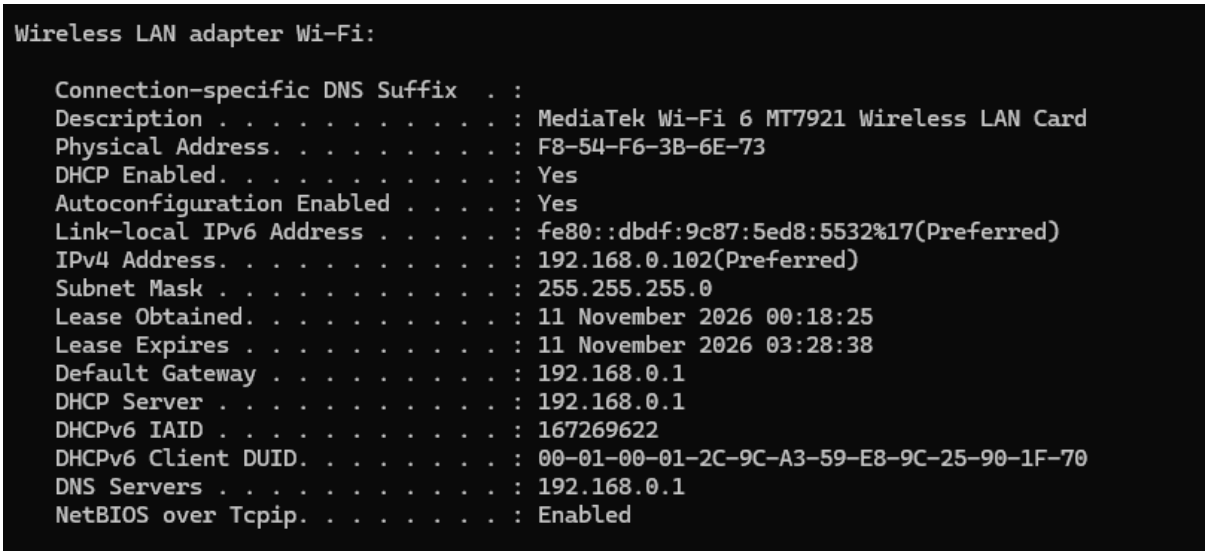


Figure 11. Module 5 – Windows ipconfig /all output showing the Wi-Fi adapter's physical address.

4.3.4 Network Topology

The Zenmap Topology view displayed the six discovered hosts around the local host node. The legend described green host markers as hosts with fewer than three open ports; however, the selected scan was Ping Scan, not a port scan. Dashed lines represented the absence of traceroute information.

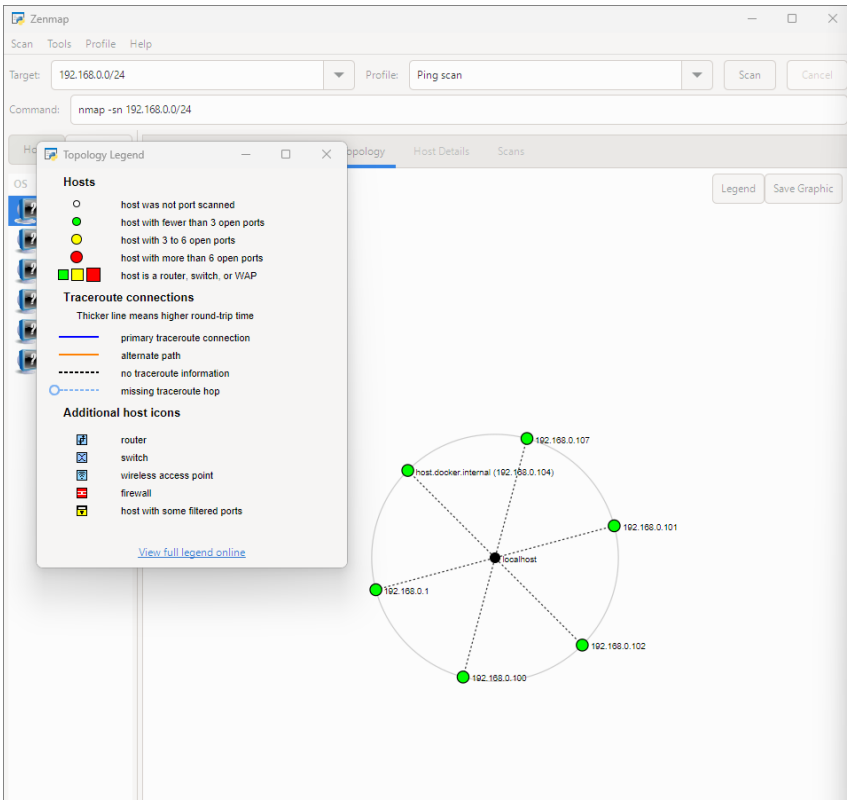


Figure 12. Module 5 – Zenmap Topology view with the topology legend visible.

The topology was exported separately as Week2_Zenmap_Network_Topology.pdf and retained as an evidence file.

5. Risk Analysis / Impact

The following are potential information-exposure or asset-management observations, not confirmed vulnerabilities. Ratings are qualitative and do not establish exploitability.

#	Observation	Potential Impact	Level
1	Public WHOIS and name-server information	Can help an observer understand domain registration and DNS delegation.	Low
2	Web technology and HTTP metadata visible	May support technology fingerprinting and further authorized review.	Low
3	Public DNS records and mail configuration	Can help map public-facing infrastructure and email routing.	Low–Medium
4	WAFW00F did not generically detect a WAF	A negative generic result does not confirm the absence of protective controls.	Informational
5	Host and IP datasets returned by theHarvester	Publicly indexed hostnames and infrastructure details may increase the external information footprint.	Medium
6	Six live hosts found on the local subnet	Unrecognized devices should be inventoried and verified by the network owner.	Low–Medium

No exploitation or vulnerability validation was performed. A hostname, IP address, DNS record, HTTP header or live host does not by itself indicate a security weakness. Further authorized assessment would be needed to establish whether a control is misconfigured or a vulnerability exists.

6. Recommendations

• Review public domain and DNS information periodically and remove records that are no longer required.

• Keep web platforms, dependencies and exposed services updated; review technology disclosures in the context of security policy.

• Review HTTP response headers and ensure browser security controls are configured appropriately.

• Maintain and monitor web application protection controls; do not infer that a WAF is absent solely from a generic WAFW00F result.

• Maintain an authoritative inventory of public subdomains and hosts, validate ownership, and investigate unexpected or obsolete entries.

• Maintain an up-to-date inventory of devices connected to the local network and investigate unknown devices.

• Restrict network scanning to owned systems or explicitly authorized scopes, and document the approved target range before testing.

• Use API credentials only where authorized and document data-source limitations when using aggregation tools such as theHarvester.

7. Conclusion

During Week 2 of the Networkwalks cybersecurity internship, I completed practical activities in domain footprinting, open-source reconnaissance and local network discovery. In Module 1, I used WHOIS, WhatWeb, nslookup, curl, WAFW00F and DNSRecon to examine public registration, web, HTTP and DNS information for github.com.

In Module 4, I used theHarvester with the Hackertarget source and then with all configured sources. The runs returned host and IP information, while several providers were limited by missing or invalid API credentials or service availability. This demonstrated the importance of understanding the sources and limitations behind automated reconnaissance tools.

In Module 5, I used Zenmap to scan my own Wi-Fi subnet, 192.168.0.0/24. The scan identified six live hosts. I recorded their IP addresses, obtained the local Windows Wi-Fi MAC address, reviewed the topology and exported it to PDF.

Overall, these exercises strengthened my understanding of reconnaissance workflows, DNS and HTTP information gathering, OSINT aggregation, network discovery and technical evidence documentation. They reinforced the importance of authorization, scope control and careful interpretation of scan results.